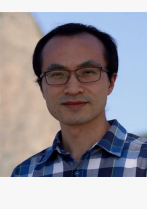


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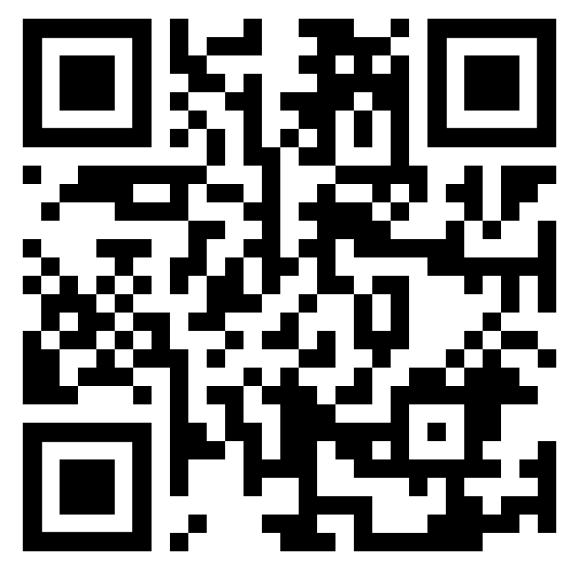
TL;DR: RapidEarth Unveiled

Imagine scanning the Earth’s surface for specific objects, not in hours, but seconds. That’s RapidEarth for you! It revolutionizes geospatial search with its innovative combination of decision branches and multi-dimensional indexing, cutting through the clutter to find the needles in the haystack instantly. Precision and speed in one package—say goodbye to the days of exhaustive searches!

Scan to check our papers

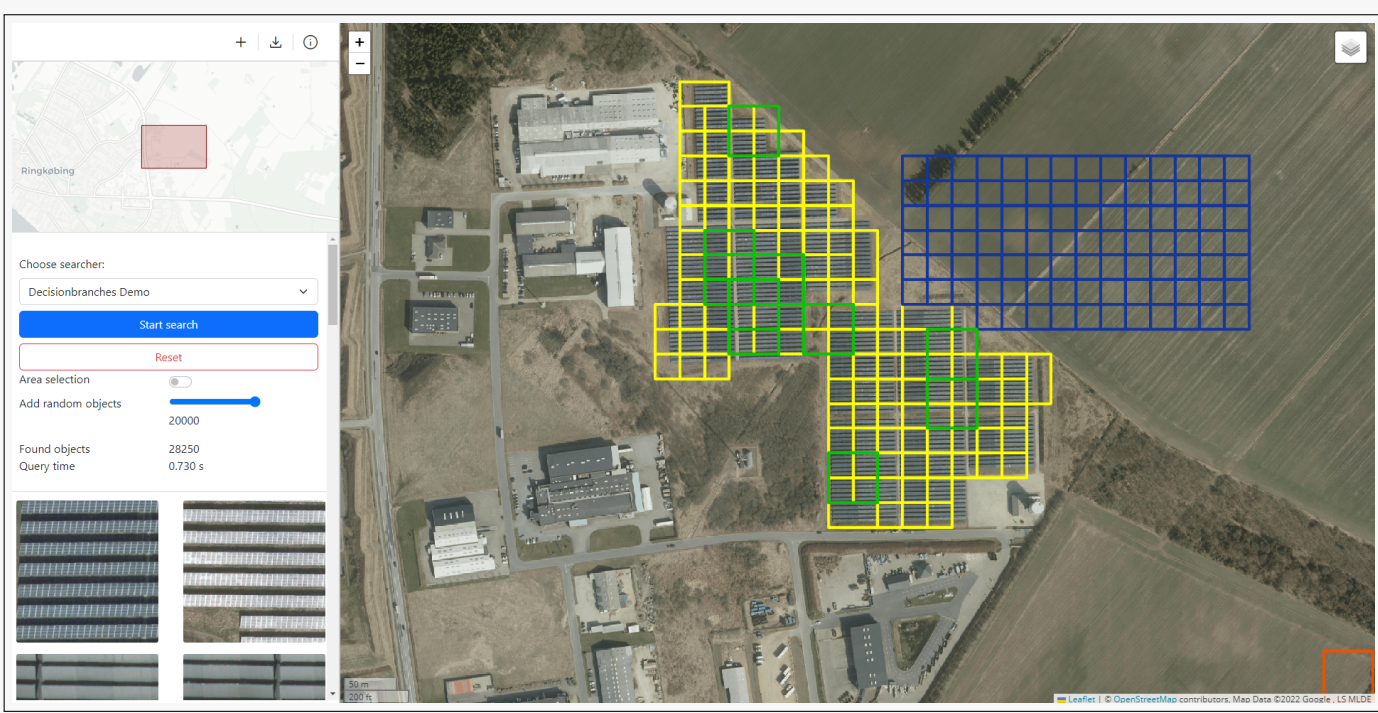


Demo Paper
SIGSPATIAL'23



Techn. Background
VLDB'23

Try out our live demo!



<https://web.rapid.earth/>

References:

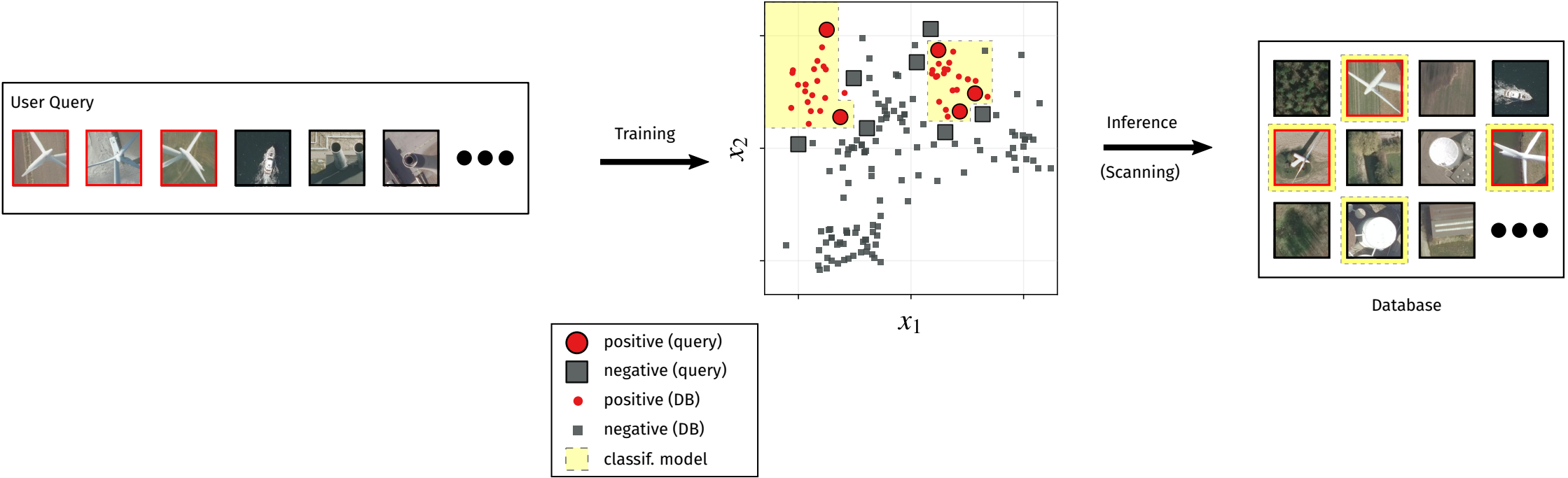
[1] C. Lülfi, D. M. L. Martins, M. A. V. Salles, Y. Zhou, and F. Gieseke. 2023. RapidEarth: A Search-by-Classification Engine for Large-Scale Geospatial Imagery. Proceedings of the ACM SIGSPATIAL (2023).
[2] C. Lülfi, D. M. L. Martins, M. A. V. Salles, Y. Zhou, and F. Gieseke. 2023. Fast Search-by-Classification for Large-Scale Databases Using Index-Aware Decision Trees and Random Forests. Proceedings of the VLDB Endowment 16 (2023).

RapidEarth: A Search-by-Classification Engine for Large-Scale Geospatial Imagery

Search-by-Classification: Vision and Problem

Imagine effortlessly pinpointing target items in vast databases using just a **handful of examples**. Examples of target objects are **easily** provided by users over satellite imagery, astronomy, and many other domains.

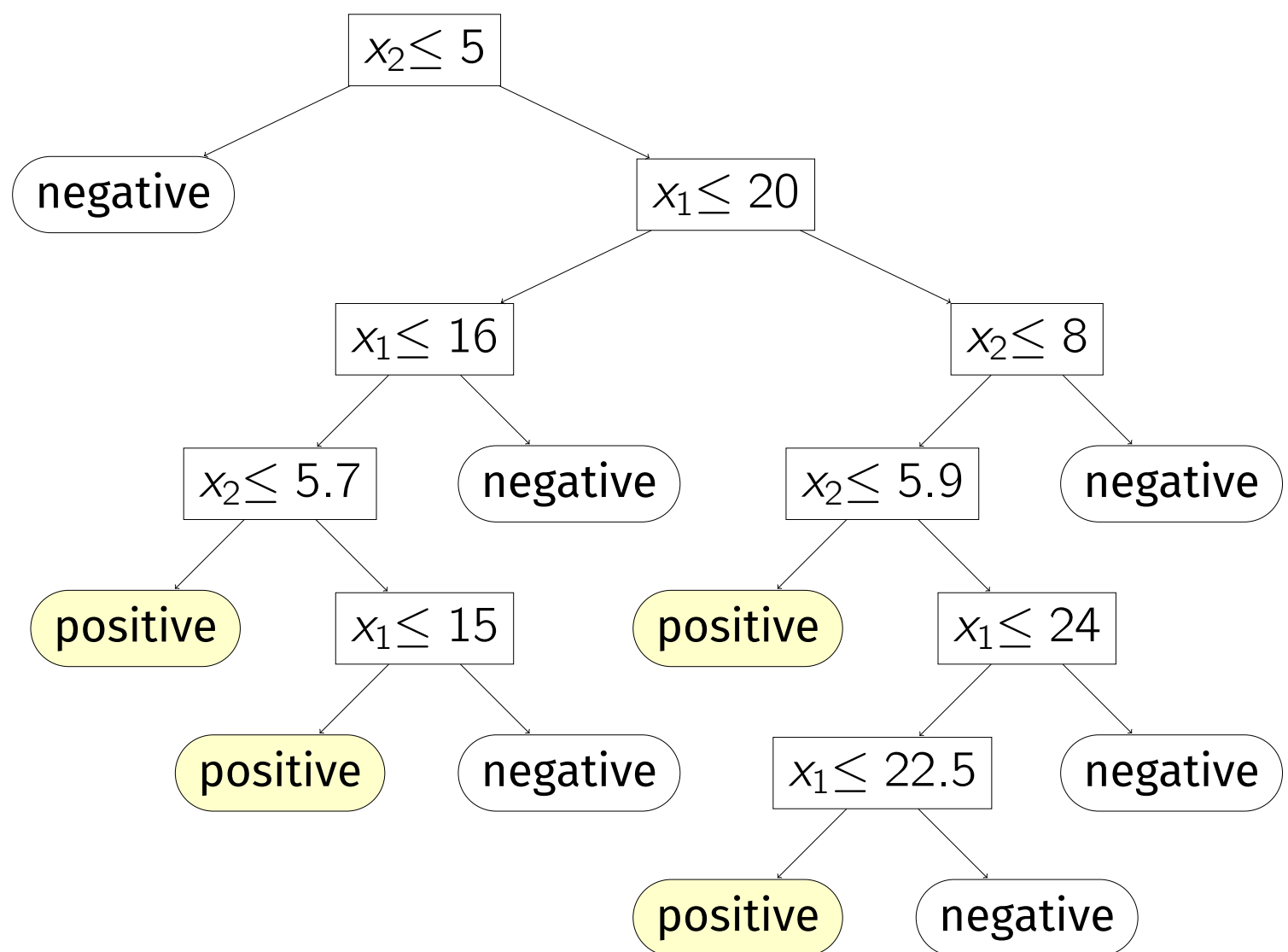
Define the user query with **positive** and **negative** examples and train an ML model with the data. **BUT** applying the learned model requires **scanning** the data. **~30 minutes** for 10^9 data points on a fast SDD.



Our idea: Use index structures to speed up inference. **No Scans needed!**

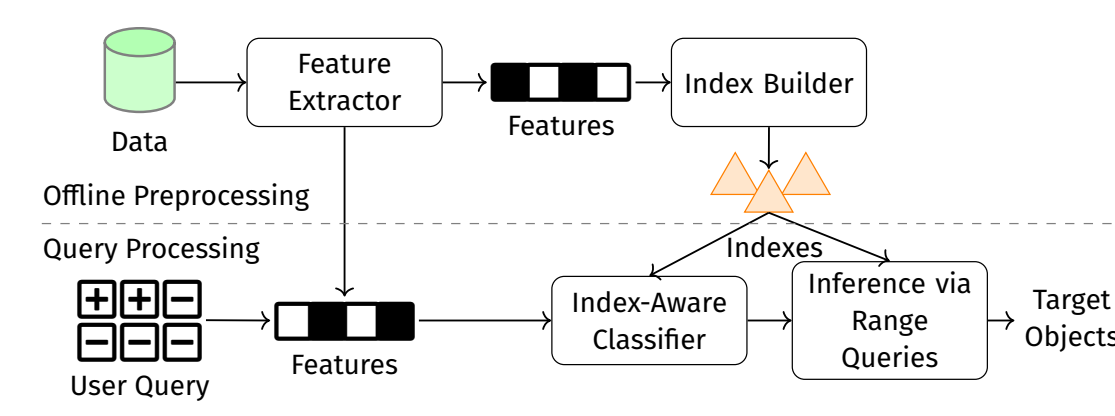
ML Model & Range Queries

Tree-based ML models can be translated into orthogonal **range queries**.

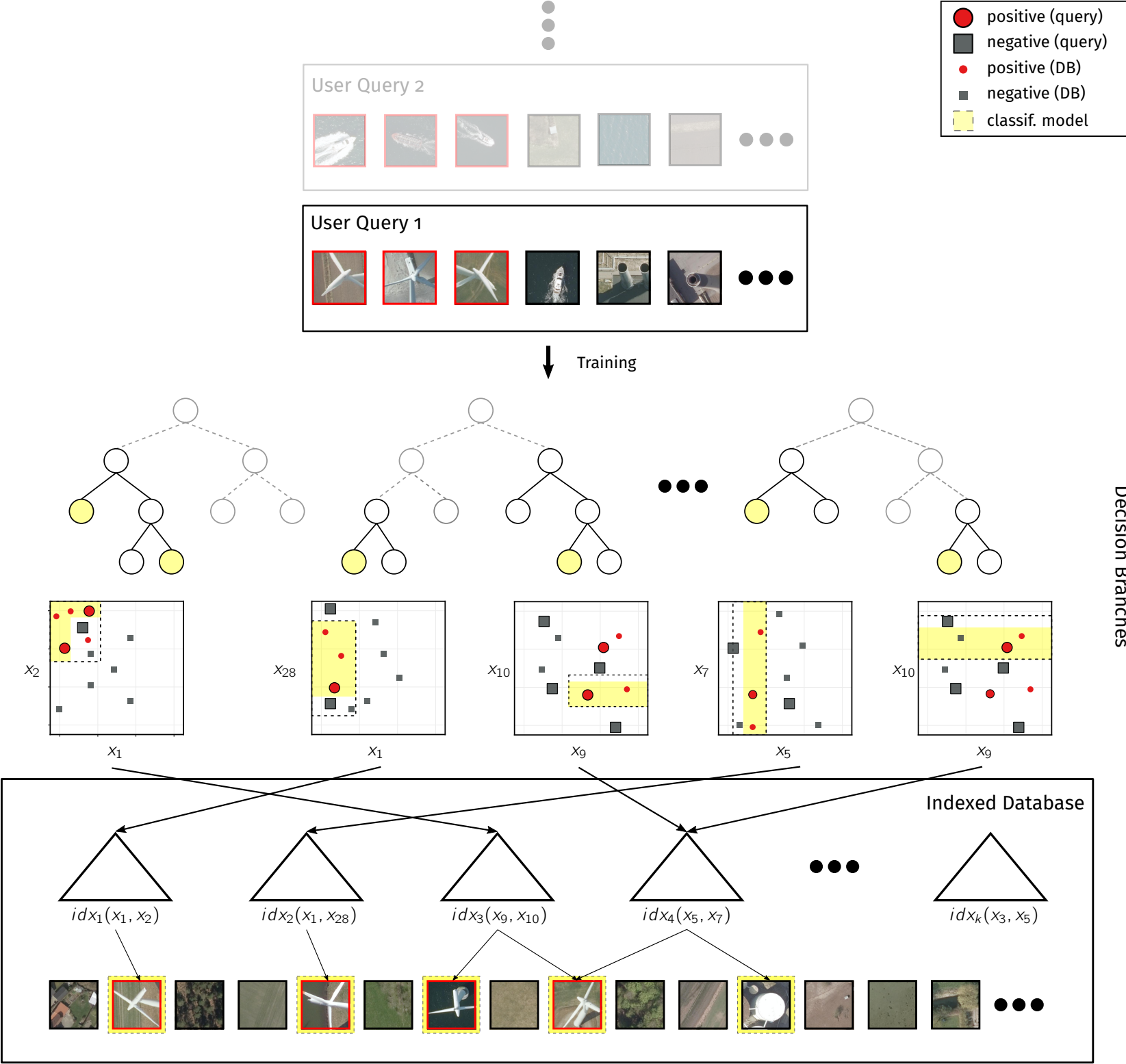


Our Approach

1. **Co-design** of index structures and machine learning model.
2. **Index-aware** construction scheme for decision trees and random forests.
3. **Similar classification performance** of regular tree-based models.
4. Trained model can be **efficiently queried** using index structures.



Decision Branches



Index-aware Classifier

Challenge: Index structures are efficient in small feature subsets; ML models usually require many features for good performance.

Observation: Using small, random feature subsets does not hurt classification performance.

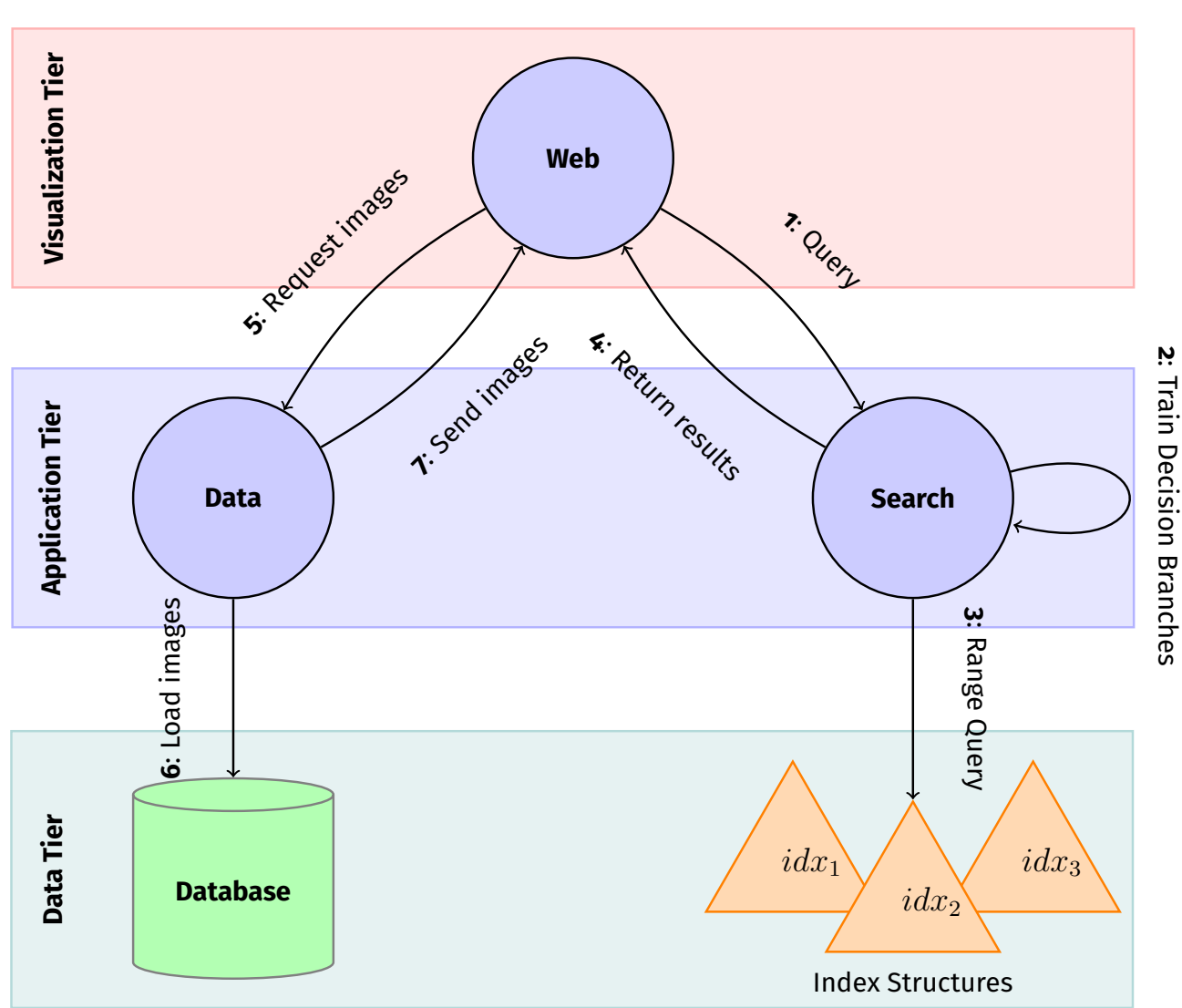
Solution:

- Index structures are prebuilt for small, random feature subsets.
- Decision Branches are trained within these feature subspaces.
- Trained Decision Branches are translated into range queries which are supported by the pre-built index structures

RapidEarth

Our demo application showcases a search-by-classification engine using an **aerial dataset of Denmark** with 12.5 cm/pixel resolution. We divided the dataset into 400x400 pixel patches with a 200-pixel step size, creating over **90 million patches**. We used **Vision Transformers** as a self-supervised feature extractor to learn intrinsic data characteristics. In total, the extracted features result into **130 GB** of storage.

System Architecture

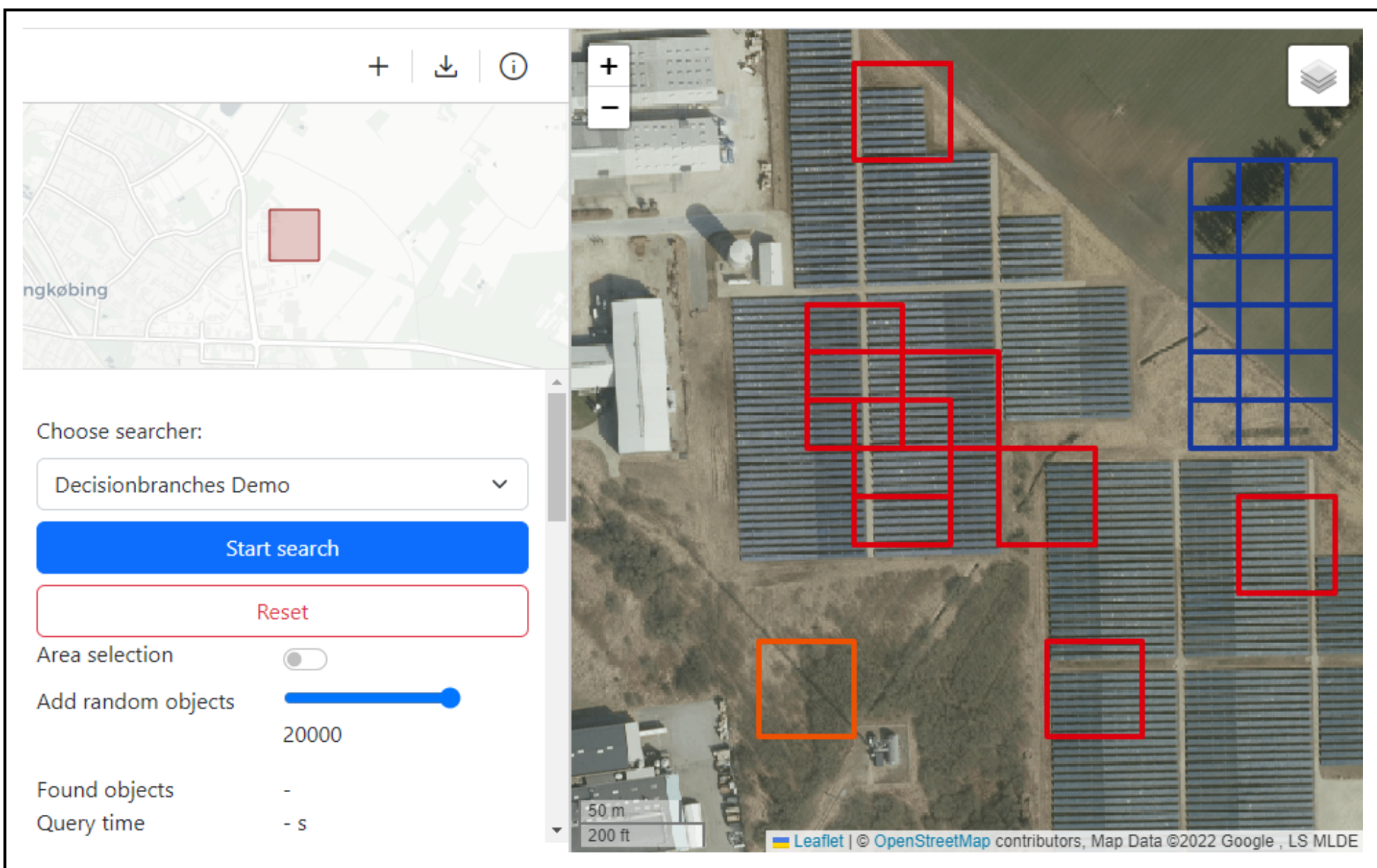


Components

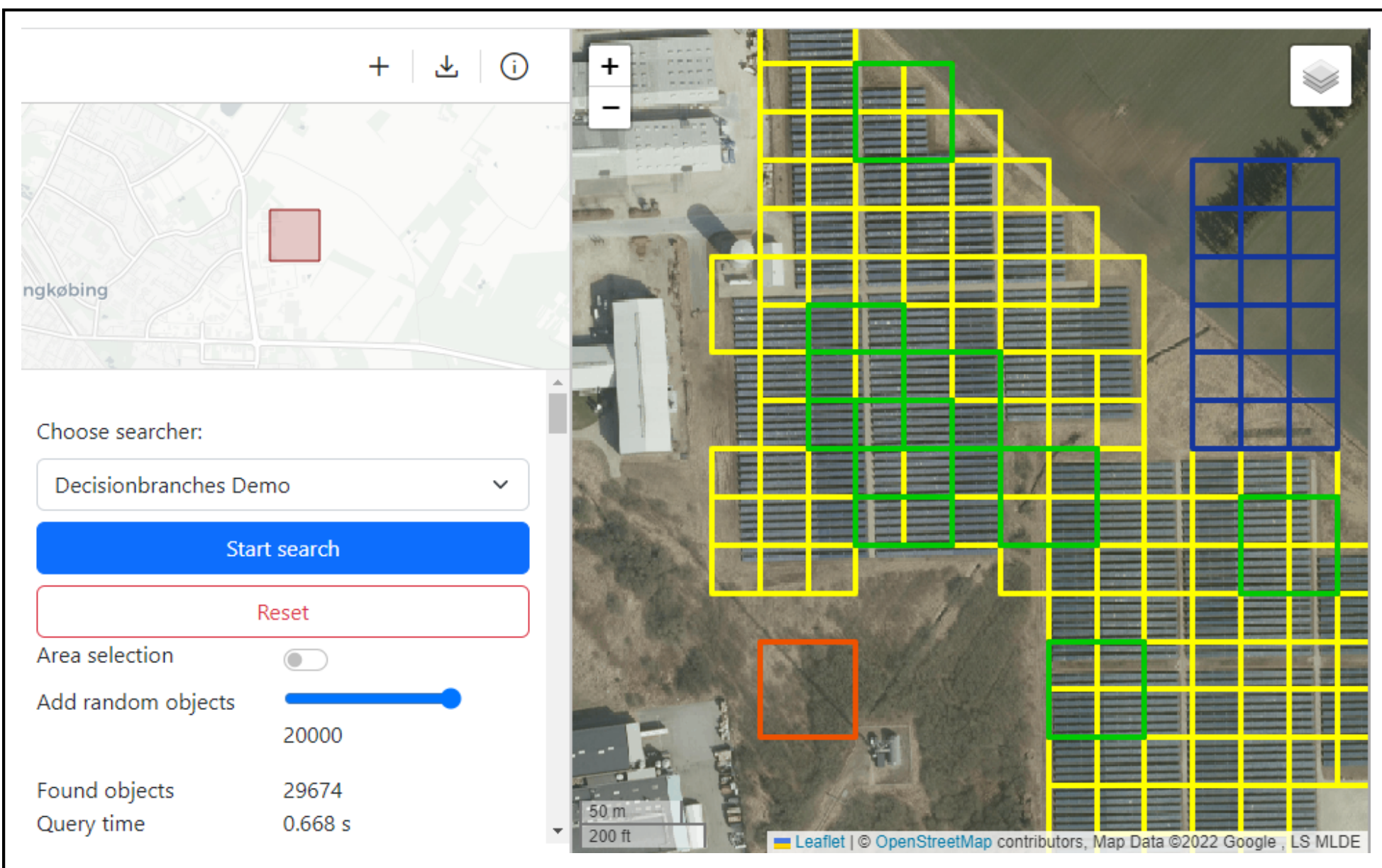
1. **Web app:** hosts the web frontend of search engine. Implemented using JavaScript and the *Leaflet* framework.
2. **Search app:** performs search-by-classification task. Retrieves database instances via range queries on index structures. Implemented with *FastAPI*.
3. **Data app:** stores underlying image patches. Implemented with *FastAPI*.

Demo

Legend: Positive selected Negative selected Location of cursor Result Result + selected



(a) Define User Query



(b) Retrieve Results